21

Nonnegative Matrices

Historical Note

Oskar Perron was working on a generalisation of the continued fraction algorithm in 1907 when he noticed that a matrix with all entries positive has a largest eigenvalue that is real, positive, simple, and equipped with an eigenvector all of whose entries are positive. It is an odd theorem to meet in that context and it turned out to matter everywhere else. Georg Frobenius spent three papers between 1908 and 1912 finding exactly how much of it survives when zeros are allowed, and the answer, that the relevant hypothesis is a connectivity condition on a directed graph, is one of the earliest places where graph theory and matrix theory are the same subject. Andrei Markov had introduced his chains in 1906, in a dispute about whether the law of large numbers required independence; he applied them to the vowels and consonants of Pushkin. The largest computation ever built on the theorem is the one Brin and Page described in 1998.

Chapter 10 asserted that a Markov chain has a stationary distribution and that its powers converge *if* every other eigenvalue is smaller in absolute value, and left the condition hanging. Chapter 9 and Chapter 19 both claimed that PageRank works. This chapter supplies the theorem underneath all three, which is about a class of matrices defined by nothing more than the signs of their entries. Throughout, $u = (1, 1, \dots, 1)$ denotes the all-ones vector, as in Chapter 19.

21.1 Two orders, and which one is meant

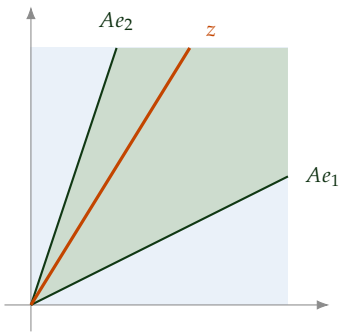
Definition 21.1 (Entrywise order)**.** *For matrices or vectors of the same shape write $A \geq B$ when $a_{ij} \geq b_{ij}$ for every i and j, and $A > B$ when the inequality is strict in every entry. A matrix is* nonnegative *if $A \geq 0$ and* positive *if $A > 0$.*

Caution

This is not the order of Chapter 15, and the clash is unavoidable because both notations are standard. There, $A \geq 0$ meant that A is symmetric with $\langle Ax, x\rangle \geq 0$ for every x; here $A \geq 0$ means every entry is at least zero. Neither implies the other. The matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ has every entry positive, but its eigenvalues are 3 and -1, so it is not positive semidefinite; the matrix $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ is positive semidefinite, with eigenvalues 2 and 0, and yet one of its entries is negative. The two symbols are also hard to tell apart in print, which is a further reason to say in words which order is meant. This chapter uses $\geq$ and $>$ entrywise and never uses the definiteness order. When reading anywhere else, check which is meant before believing a deduction.

Definition 21.2 (Spectral radius). $\rho(A) = \max\{|\lambda| : \lambda \in \operatorname{Spec}(A)\}$, *as in Chapter 18.*

The question this chapter answers is what nonnegativity alone forces about $\rho(A)$ and its eigenvector. The answer is a good deal, and the reason is geometric: a nonnegative matrix maps the nonnegative cone into itself, and a positive matrix maps it strictly inside itself.



the pale square is the nonnegative cone and the green wedge is its image; the ray z is the one direction the map leaves alone

Figure 21.1. Why Perron's theorem is true, for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. Because every entry is positive, A carries the nonnegative quadrant into a wedge that meets the boundary only at the origin. Iterating squeezes the wedge further, and the directions it closes down on form the single ray spanned by the Perron vector $z = (1, 1.618)$, with $Az = \rho z$ and $\rho = \frac{1}{2}(5 + \sqrt{5}) \approx 3.618$. The proof of Theorem 21.5 is this picture made arithmetic.

21.2 Irreducibility

Definition 21.3 (Irreducible, primitive)**.** *Let $A \geq 0$ be $n \times n$ with $n \geq 2$. Its* digraph *has vertices $1, \dots, n$ and an edge $i \to j$ whenever $a_{ij} > 0$. Call A* irreducible *if that digraph is strongly connected, that is, if every vertex can be reached from every other by a directed path. Call A* primitive *if $A^k > 0$ for some $k \geq 1$.*

Since $(A^k)_{ij}$ is a sum of nonnegative products taken over the directed paths of length k from i to j, it is positive exactly when at least one such path exists. So irreducibility says that for each pair (i, j) some power of A has a positive (i, j) entry, and primitivity says that one single power works for all pairs at once. Every positive matrix is primitive, with $k = 1$, and every primitive matrix is irreducible.

Lemma 21.4 (Irreducibility made positive)**.** *If $A \geq 0$ is irreducible then $(\mathbf{1} + A)^{n-1} > 0$.*

Proof. By the binomial theorem, which applies because $\mathbf{1}$ and A commute, $(\mathbf{1} + A)^{n-1} = \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} A^\ell$, a sum of nonnegative terms. So it suffices to find, for each pair (i, j), one ℓ in that range with $(A^\ell)_{ij} > 0$. For $i = j$ take $\ell = 0$, where $A^0 = \mathbf{1}$ contributes 1. For $i \neq j$, strong connectivity gives a directed path from i to j; a shortest one visits no vertex twice, so it passes through at most n vertices and has length ℓ with $1 \leq \ell \leq n - 1$. That path witnesses $(A^\ell)_{ij} > 0$. $\square$

21.3 Perron and Frobenius

Theorem 21.5 (Perron 1907, Frobenius 1912)**.** *Let $A \geq 0$ be irreducible. Then*

(a) $\rho = \rho(A)$ is positive and is an eigenvalue of A, called the Perron root*;*

(b) some eigenvector z for ρ satisfies $z > 0$;

(c) ρ is a simple root of the characteristic polynomial, and up to scaling z is the only nonnegative eigenvector of A;

(d) $$\rho \;=\; \max_{x \geq 0,\; x \neq \mathbf{0}} \; \min_{i \,:\, x_i > 0} \frac{(Ax)_i}{x_i}.$$

Proof. Write $B = (\mathbf{1} + A)^{n-1}$, which is positive by Lemma 21.4 and commutes with A, being a polynomial in it. Let $\Delta = \{x \geq 0 : \sum_i x_i = 1\}$, which is compact, and for $x \in \Delta$ put

$$r(x) \;=\; \max\{s \geq 0 : Ax \geq sx\} \;=\; \min_{i \,:\, x_i > 0} \frac{(Ax)_i}{x_i},$$

the two descriptions agreeing because the constraint is vacuous at coordinates where $x_i = 0$. Note that r is unchanged by positive scaling, so nothing is lost by restricting to Δ.

Step 1: the supremum of r is attained. The function r is not continuous on Δ, since the set of indices in the minimum can jump. Let $\Gamma = \{Bx/\|Bx\|_1 : x \in \Delta\}$, the continuous image of a compact set and hence compact, every element of which is strictly positive because $B > 0$. On strictly positive vectors r is a minimum of finitely many continuous functions and so is continuous, so r attains a maximum on Γ. Moreover r does not decrease under $x \mapsto Bx$: applying the nonnegative matrix B to the inequality $Ax \geq r(x)x$ preserves it, giving $BAx \geq r(x)Bx$, which is $A(Bx) \geq r(x)(Bx)$ because A and B commute. Hence the supremum of r over Δ equals its maximum over Γ. Call it ρ^* and let $z \in \Gamma$ attain it, so $z > 0$.

Step 2: $Az = \rho^* z$ and $\rho^* > 0$. Put $w = Az - \rho^* z$, which is nonnegative by the definition of r. If $w \neq \mathbf{0}$ then $Bw > 0$, and since A and B commute this reads $A(Bz) - \rho^*(Bz) > 0$, so $r(Bz) > \rho^*$, contradicting maximality. So $w = \mathbf{0}$. For positivity, take $x = u/n \in \Delta$: then $r(x)$ is the smallest row sum of A, and no row sum can vanish, since a zero row would leave its vertex with no outgoing edge and destroy strong connectivity. Hence $\rho^* \geq r(u/n) > 0$.

Step 3: $\rho^* = \rho(A)$. Let $Av = \lambda v$ with $v \neq \mathbf{0}$, possibly complex, and let $|v|$ be the vector of moduli. Then for each i,

$$|\lambda|\,|v_i| \;=\; \Big|\sum_j a_{ij} v_j\Big| \;\leq\; \sum_j a_{ij}\,|v_j| \;=\; (A\,|v|)_i,$$

so $A\,|v| \geq |\lambda|\,|v|$ and therefore $|\lambda| \leq r(|v|) \leq \rho^*$. Thus $\rho(A) \leq \rho^*$; and ρ^* is itself an eigenvalue by Step 2, so the two are equal. Write ρ for the common value. Steps 1 to 3 prove (a), (b) and (d).

Step 4: the eigenspace for ρ is one-dimensional. Let $Ay = \rho y$ with y real; this is no loss, since A and ρ are real and so the real and imaginary parts of any eigenvector are themselves eigenvectors or zero. Replacing y by $-y$ if necessary, assume some $y_i > 0$, and set $t = \min\{z_i/y_i : y_i > 0\}$, which is positive because $z > 0$. Then $w = z - ty$ is nonnegative, with $w_k = 0$ at a minimising index k, and $Aw = \rho w$. If $w \neq \mathbf{0}$ then $Bw > 0$; but $Aw = \rho w$ forces $Bw = (1+\rho)^{n-1} w$, whose kth entry is zero. So $w = \mathbf{0}$ and y is a multiple of z.

For the claim that z is the *only* nonnegative eigenvector, note first that A^t is nonnegative and irreducible, since reversing every edge of a strongly connected digraph leaves it strongly connected, and has the same spectral radius as A by Proposition 9.18. So Steps 1 to 3 applied to A^t produce a *left* Perron vector $w > 0$ with $A^t w = \rho w$. Now let $Ay = \mu y$ with $y \geq 0$ and $y \neq \mathbf{0}$. Choosing i with $y_i > 0$ gives $\mu y_i = (Ay)_i \geq 0$, so $\mu \geq 0$; hence $By = (1+\mu)^{n-1} y$ with a positive factor, and $By > 0$ forces $y > 0$. Finally

$$\mu\,(w^t y) \;=\; w^t A y \;=\; (A^t w)^t y \;=\; \rho\,(w^t y),$$

and $w^t y > 0$ because both vectors are positive, so $\mu = \rho$ and the first part applies.

Step 5: ρ is algebraically simple. Let $D = \operatorname{diag}(z_1, \dots, z_n)$, which is invertible since $z > 0$, and put $C = \rho^{-1} D^{-1} A D$. Then $C \geq 0$ and

$$(Cu)_i \;=\; \frac{1}{\rho} \sum_j \frac{a_{ij} z_j}{z_i} \;=\; \frac{(Az)_i}{\rho\, z_i} \;=\; 1,$$

so every row of C sums to 1. Products of such matrices are again such matrices, so $C^k \geq 0$ with $C^k u = u$ for every k, and therefore every entry of C^k lies in $[0,1]$: the powers of C are bounded. Now C is similar to $\rho^{-1} A$, so the eigenvalue 1 of C has the same multiplicities as ρ has for A, and by Step 4 its eigenspace is one-dimensional. If its algebraic multiplicity exceeded one, the Jordan form of Theorem 11.9 would contain a block $J_m(1)$ with $m \geq 2$, whose kth power has the entry k just above the diagonal by Example 11.14; the powers of C would then be unbounded. So the multiplicity is one, which with Step 4 completes (c). □

Theorem 21.6 (Primitivity gives strict dominance)**.** *Let $A \geq 0$ be primitive. Then every eigenvalue $\lambda \neq \rho$ satisfies $|\lambda| < \rho$.*

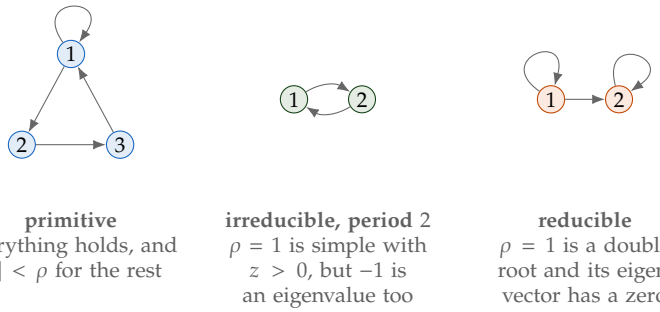
Proof. First take $A > 0$. Suppose $Av = \lambda v$ with $|\lambda| = \rho$ and $v \neq \mathbf{0}$. As in Step 3 above, $A\,|v| \geq \rho\,|v|$; if the difference w were nonzero then $Aw > 0$, giving $r(A\,|v|) > \rho$ and contradicting Theorem 21.5(d). So $A\,|v| = \rho\,|v|$, and by part (c) the vector $|v|$ is a positive multiple of z; in particular $|v| > 0$. Now the chain of inequalities in Step 3 is an equality throughout, so for each i,

$$\left| \sum_j a_{ij} v_j \right| \;=\; \sum_j a_{ij} \left| v_j \right|.$$

Equality in the triangle inequality for complex numbers, with every $a_{ij} > 0$ and every $\left|v_j\right| > 0$, forces all the v_j to share one argument, say $v = e^{i\theta}\,|v|$. Substituting into $Av = \lambda v$ and cancelling $e^{i\theta}$ gives $A\,|v| = \lambda\,|v|$, which combined with $A\,|v| = \rho\,|v|$ gives $\lambda = \rho$.

Now let A be primitive with $A^m > 0$. Triangularising as in Theorem 14.5 shows that the eigenvalues of A^m, with multiplicity, are the mth powers of those of A; in particular $\rho(A^m) = \rho^m$. Suppose $|\lambda| = \rho$ with $\lambda \neq \rho$. Then λ^m is an eigenvalue of A^m of modulus ρ^m, so by the case already proved $\lambda^m = \rho^m$. But then ρ^m arises at least twice in the list of eigenvalues of A^m, once from ρ and once from λ, contradicting the algebraic simplicity of $\rho(A^m)$ given by Theorem 21.5(c) applied to the positive matrix A^m. □

Figure 21.2 shows why each hypothesis is there, one digraph at a time. The middle matrix is irreducible but not primitive, and its powers alternate for ever rather than converging.



primitive
everything holds, and $|\lambda| < \rho$ for the rest

irreducible, period 2
$\rho = 1$ is simple with $z > 0$, but -1 is an eigenvalue too

reducible
$\rho = 1$ is a double root and its eigenvector has a zero

Figure 21.2. Each hypothesis earns its place. The middle matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which is irreducible, so Theorem 21.5 applies in full, but is not primitive, and indeed its powers alternate rather than converge: Theorem 21.6 is exactly what fails. The right-hand matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, where vertex 1 cannot be reached from vertex 2; here even simplicity fails, since $\rho = 1$ has a Jordan block of size two, and the only eigenvector is $(1, 0)$.

Corollary 21.7 (Row sum bounds)**.** *For $A \geq 0$ irreducible, the Perron root lies between the smallest and the largest row sum.*

Proof. The lower bound is Theorem 21.5(d) evaluated at $x = u$, where $r(u) = \min_i \sum_j a_{ij}$. For the upper bound, if $Av = \lambda v$ pick i with $|v_i|$ largest; then $|\lambda||v_i| \leq \sum_j a_{ij}\,|v_j| \leq \left(\sum_j a_{ij}\right)|v_i|$, so $|\lambda|$ is at most the ith row sum. □

21.4 Markov chains and mixing

Definition 21.8 (Stochastic matrix)**.** $P \geq 0$ *is* column-stochastic *if every column sums to* 1*, that is $u^{t}P = u^{t}$. A* probability vector *is $x \geq 0$ with $u^{t}x = 1$.*

A column-stochastic P maps probability vectors to probability vectors, which is what makes $x \mapsto Px$ one step of a chain. Its spectral radius is exactly 1: if $Px = \lambda x$ then

$$|\lambda| \sum_i |x_i| \;=\; \sum_i |(Px)_i| \;\leq\; \sum_i \sum_j p_{ij}\,|x_j| \;=\; \sum_j \Big(\sum_i p_{ij}\Big)\,|x_j| \;=\; \sum_j |x_j|,$$

so $|\lambda| \leq 1$; and 1 is attained because $u^{t}P = u^{t}$ makes 1 an eigenvalue of P^{t}, hence of P by Proposition 9.18.

Theorem 21.9 (Stationary distributions). *Let P be column-stochastic and irreducible. Then there is exactly one probability vector π with $P\pi = \pi$, and $\pi > 0$. If in addition P is primitive then*

$$P^k \longrightarrow \pi\, u^t, \qquad \text{so} \qquad P^k x_0 \longrightarrow \pi \ \text{ for every probability vector } x_0,$$

and the error decays like $|\lambda_2|^k$, where λ_2 is an eigenvalue of second largest modulus.

Proof. By the display above $\rho(P) = 1$, so Theorem 21.5 applies with $\rho = 1$: it is a simple eigenvalue with a positive eigenvector, unique up to scale, and scaling it to sum to 1 gives π. Uniqueness among probability vectors is the simplicity.

For the limit, let $E = \pi u^t$. Then $E^2 = \pi(u^t\pi)u^t = E$ since $u^t\pi = 1$, and $PE = (P\pi)u^t = E$ while $EP = \pi(u^tP) = \pi u^t = E$. So E is idempotent, commutes with P, has image the line spanned by π, and has kernel $\{x : u^t x = 0\}$.

That kernel is exactly the sum W of the generalised eigenspaces of the other eigenvalues: if $(P - \lambda\mathbf{1})^m y = \mathbf{0}$ with $\lambda \neq 1$ then applying u^t and using $u^tP = u^t$ gives $(1-\lambda)^m\, u^t y = 0$, hence $u^t y = 0$; and $\dim W = n - 1$ because 1 is simple, which matches $\dim \ker E$. So E is the spectral projection belonging to the eigenvalue 1.

Put $R = P - E$. Then $RE = ER = 0$, so $P^k = E + R^k$, and the eigenvalues of R are 0 together with the eigenvalues of P other than 1, all of modulus at most $|\lambda_2| < 1$ by Theorem 21.6. Writing R in Jordan form by Theorem 11.9 and using the formula for the powers of a block from Example 11.14, every entry of R^k is a binomial multiple of λ^{k-j} with $|\lambda| < 1$, so $R^k \to 0$. Finally $Ex_0 = \pi(u^t x_0) = \pi$. □

Key Idea

That is the missing piece of Chapter 10. The Insight there said that a Markov matrix "always has 1 as an eigenvalue, with the stationary distribution as its eigenvector, and if every other eigenvalue is smaller in absolute value then A^k converges". Every clause is now a theorem, and the hypothesis that looked like a technicality has a name and a meaning: the chain must be able to get from any state to any other, and it must not be trapped in a cycle. The spectral gap $1 - |\lambda_2|$ is the mixing rate, exactly as claimed.

Caution

Two honest qualifications. If λ_2 is a defective eigenvalue, its Jordan block contributes a polynomial factor and the decay is $k^{m-1}\,|\lambda_2|^k$ rather than $|\lambda_2|^k$; the conclusion is the same but the constant is worse. And $|\lambda_2|$ governs the *asymptotic* rate only. For a

chain with a small gap the transient behaviour before the asymptotics take hold can be long, which is why the practical study of mixing times does not stop at the second eigenvalue.

Example 21.10 (Two states, exactly). *Let* $P = \left(\begin{smallmatrix} 1-a & b \\ a & 1-b \end{smallmatrix}\right)$ *with* $0 < a, b < 1$, *so both columns sum to* 1 *and* $P > 0$, *hence primitive. Its trace is* $2 - a - b$ *and its determinant is* $(1-a)(1-b) - ab = 1 - a - b$, *so the eigenvalues are* 1 *and* $1 - a - b$. *A short check gives*

$$\pi = \frac{1}{a+b}\begin{pmatrix} b \\ a \end{pmatrix}, \qquad \text{spectral gap} = 1 - |1 - a - b|\,.$$

With $a = b = \tfrac{1}{2}$ *the second eigenvalue is* 0 *and the chain reaches* π *in one step. With* $a = b = 0.01$ *it is* 0.98, *and the distance to* π *halves every* $\ln 2/\ln(1/0.98) \approx 34$ *steps. The stationary distribution is unchanged by that; only the time to reach it is different.*

21.5 PageRank

The link matrix of the web is column-stochastic once dangling pages are patched, but it is very far from irreducible: most of the web cannot be reached from most of it. So Theorem 21.9 does not apply, and neither existence nor uniqueness of the ranking is available. The fix is one line of linear algebra.

Proposition 21.11 (Damping). *Let* S *be column-stochastic, let* v *be a probability vector and let* $0 < \alpha < 1$. *Then*

$$G \;=\; \alpha S + (1-\alpha)\, v\, u^{t}$$

is column-stochastic; it is positive whenever $v > 0$; *and every eigenvalue of* G *other than* 1 *satisfies* $|\lambda| \le \alpha$.

Proof. For the first claim, $u^{t}G = \alpha u^{t}S + (1-\alpha)(u^{t}v)u^{t} = \alpha u^{t} + (1-\alpha)u^{t} = u^{t}$, using $u^{t}v = 1$. Positivity is clear since $(1-\alpha)vu^{t} > 0$ when $v > 0$.

For the third, let $Gx = \lambda x$ with $\lambda \neq 1$. Applying u^{t} and using $u^{t}G = u^{t}$ gives $\lambda\, u^{t}x = u^{t}x$, so $u^{t}x = 0$. But then $v(u^{t}x) = \mathbf{0}$, so $\lambda x = Gx = \alpha S x$, that is $Sx = (\lambda/\alpha)x$. Hence λ/α is an eigenvalue of S, and $|\lambda|/\alpha \le \rho(S) = 1$. □

Key Idea

The damping term does two jobs with one stroke. It makes G positive, so Theorem 21.5 guarantees that the ranking exists, is unique and is strictly positive, no matter what the link structure looks like; and it puts a hard bound α on the second eigenvalue, so the power method of Chapter 9 converges at a rate that does not depend on the web at all. With $\alpha = 0.85$, reaching 10^{-8} takes $8 \ln 10 / \ln(1/0.85) \approx 114$ iterations, each of which is one sparse matrix-vector product. That is the whole algorithm, and the only free parameter trades ranking quality against the number of iterations: at $\alpha = 0.99$ the same accuracy needs about 1830.

Figure 21.3 shows the damping doing its work on the spectrum. Before it, several eigenvalues sit on the unit circle and the power method does not converge; after it, one eigenvalue is at 1 and the rest are inside a circle of radius α.

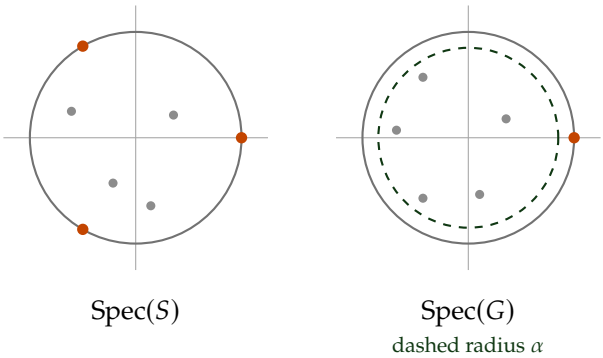


Figure 21.3. Proposition 21.11 drawn. On the left a column-stochastic matrix whose digraph has a periodic component: several eigenvalues sit on the unit circle, so the power method does not converge and the stationary vector is not unique. On the right the damped matrix, whose only eigenvalue of modulus 1 is the Perron root and whose remaining spectrum is confined to the disc of radius α. The gap between the two circles is what the power method converges at, and it was put there deliberately.

21.6 Nonnegative matrix factorisation

Given $A \geq 0$ of size $m \times n$ and a target rank r, nonnegative matrix factorisation asks for $W \geq 0$ of size $m \times r$ and $H \geq 0$ of size $r \times n$ minimising $\|A - WH\|_F$. Since the columns of WH are nonnegative combinations of the columns of W, nothing cancels, and the columns of W read as additive parts of which the data is built. That is the entire appeal, and it is a real one.

Proposition 21.12 (Nonnegative rank). *Let* $\operatorname{rank}_+(A)$ *be the smallest* r *for which* $A = WH$ *with* $W, H \geq 0$ *exactly. Then* $\operatorname{rank}_+(A) \geq \operatorname{rank}(A)$.

Proof. If $A = WH$ with W of r columns then $\operatorname{rank} A = \operatorname{rank}(WH) \leq \operatorname{rank} W \leq r$ by Exercise 6.29, and taking $r = \operatorname{rank}_+(A)$ gives the claim. □

The inequality can be strict, which is the first sign that nonnegative rank is a harder invariant than rank. The standard small example is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix},$$

whose rows satisfy $r_1 - r_2 - r_3 + r_4 = \mathbf{0}$ while any three of them are independent, so $\operatorname{rank} M = 3$. Its nonnegative rank is 4; that lower bound is not obvious and we take it from Cohen and Rothblum [152].

Insight

Perron's theorem explains why the singular value decomposition cannot do this job. If $A \geq 0$ then $A^t A \geq 0$, so when it is irreducible Theorem 21.5 supplies a nonnegative leading eigenvector v_1, and $u_1 = Av_1/\sigma_1 \geq 0$: the *first* singular pair of a nonnegative matrix can always be taken nonnegative. But v_2 must be orthogonal to $v_1 > 0$, and a nonzero vector orthogonal to a strictly positive vector must have entries of both signs. So every singular vector after the first necessarily mixes signs, and the rank-r approximation of Theorem 16.12 necessarily represents the data by cancellation. That is not a defect of any algorithm. It is forced by orthogonality, and it is precisely the gap that nonnegative factorisation exists to fill.

Caution

What nonnegative factorisation gives up is everything Chapter 16 guaranteed. The optimum need not be unique, even up to the obvious rescaling $W \mapsto WD$, $H \mapsto D^{-1}H$. Computing it is NP-hard, so the standard multiplicative-update and alternating least squares algorithms find local minima and different initialisations give different parts. There is no analogue of Eckart and Young, so a factorisation at rank $r + 1$ need not contain the one at rank r. The honest summary is the mirror image of the one in Chapter 19 about low-rank adaptation: the singular value decomposition is optimal but signed, and nonnegative factorisation is interpretable

but neither optimal nor well-posed. Which defect matters is a question about the application, not about the mathematics.

Open Problem 21.13 (The eigenvalues of a doubly stochastic matrix)**.** *Let $\Theta_n \subset \mathbb{C}$ be the set of all complex numbers that arise as an eigenvalue of some $n \times n$ doubly stochastic matrix.* What is Θ_n?

What is known. *Very little, and less than was believed. Every doubly stochastic matrix has 1 as an eigenvalue and all eigenvalues in the closed unit disc, by Chapter 21. The permutation matrices contribute every root of unity of order at most n, and convexity plus the Birkhoff theorem gives that Θ_n contains the union of the polygons with those roots of unity as vertices. Perfect and Mirsky conjectured in 1965 that Θ_n is exactly that union.* The conjecture is true for $n \le 3$ and false for $n = 5$, *by an explicit counterexample of Mashreghi and Rivard [456]. What Θ_n actually is, for any $n \ge 4$, is unknown.*

Where the difficulty sits. *Birkhoff's theorem says the doubly stochastic matrices are the convex hull of the permutation matrices, so the set is as well understood as a polytope can be. But the eigenvalue map is not convex, and the image of a convex set under a nonconvex map need not be anything nice. The counterexample at $n = 5$ was found by pushing a parameterised family to the boundary of the conjectured region and going past it, which is the only technique available.*

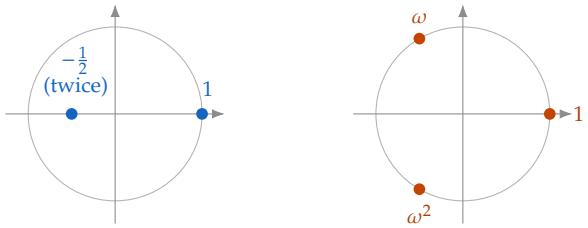
Where a reader might start. *Compute Θ_2 completely: a 2×2 doubly stochastic matrix is $\left(\begin{smallmatrix} t & 1-t \\ 1-t & t \end{smallmatrix}\right)$, so its eigenvalues are 1 and $2t - 1$, and $\Theta_2 = [-1, 1]$. Then do $n = 3$ by parameterising the Birkhoff polytope, which has six vertices, and drawing the region numerically. The picture that emerges, a triangle-like region with vertices at the cube roots of unity, is exactly what Perfect and Mirsky generalised, and seeing it makes the failure at $n = 5$ genuinely surprising.*

Example 21.14 (Primitive and imprimitive, side by side)**.** *Take the adjacency matrix of the triangle, $A = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$. Every row sums to 2, so $A\mathbf{1} = 2 \cdot \mathbf{1}$ and the Perron root is 2 with eigenvector $(1, 1, 1)$, positive as Theorem 21.5 promises. The other eigenvalues are -1 twice, both of modulus strictly below 2, so powers of $A/2$ converge and the walk mixes. Now take the cyclic shift $B = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. It is nonnegative and irreducible, so Perron and Frobenius still applies and the spectral radius is 1. But the eigenvalues are the three cube roots of unity,* all of modulus one. *Nothing converges: $B^3 = \mathbf{1}$, and a walker returns to the start every three steps and never spreads out.*

The difference is primitivity. For A, $(\mathbf{1} + A)^2$ is strictly positive; for B, $(\mathbf{1} + B)^2 = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 1 \end{smallmatrix}\right)$

is positive too, yet B^k never is, because B has period three. Irreducibility gives you a positive Perron vector; only primitivity gives you convergence, *and Lemma 21.4 tests the first, not the second.*

Figure 21.4 draws the two spectra behind that sentence: for the primitive matrix the other eigenvalues sit strictly inside the unit circle, and for the periodic one they sit on it.



triangle: one eigenvalue on the circle, the rest strictly inside

three-cycle: every eigenvalue on the circle

Figure 21.4. Irreducibility gives a positive Perron vector; only primitivity gives convergence. Both matrices here are nonnegative and irreducible with spectral radius 1 after scaling, so Perron and Frobenius applies to each. On the left, the triangle's other eigenvalues sit strictly inside the unit circle, so powers converge and a random walk mixes. On the right, the cyclic shift has all three eigenvalues *on* the circle, at the cube roots of unity; nothing converges, $B^3 = \mathbf{1}$, and a walker returns to its start every three steps forever. Lemma 21.4 tests irreducibility, which both pass, and says nothing about the distinction drawn here. Example 21.14 gives the matrices.

Exercises

Exercise 21.15. *Decide, for each of $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, say whether it is nonnegative in the sense of Definition 21.1 and whether it is positive semidefinite in the sense of Chapter 15. One of the three is both, one is only the first and one is only the second; identify which is which, and then give a fourth matrix that is neither.*

Exercise 21.16. *For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ compute the Perron root and a Perron vector exactly, confirm the picture in Figure 21.1, and check Corollary 21.7. Then evaluate the expression in Theorem 21.5(d) at $x = (1, 0)$, $x = (1, 1)$ and $x = (1, 2)$, and observe that all three are at most ρ.*

Exercise 21.17. *Draw the digraph of* $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ *and decide whether it is irreducible and whether it is primitive. Compute* $(\mathbf{1} + A)^2$ *and check Lemma 21.4. Find the smallest* k *with* $A^k > 0$.

Exercise 21.18. *Let* $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. *Verify every conclusion of Theorem 21.5 for it, then show that* A^k *does not converge. Which step of the proof of Theorem 21.6 fails, and at which matrix entry?*

Exercise 21.19. *Let* $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. *Show that* $A \geq 0$ *but is not irreducible, that* $\rho(A) = 1$ *has algebraic multiplicity two, and that no positive eigenvector exists. Which hypothesis of Theorem 21.5 is being used where, and where does Lemma 21.4 break down?*

Exercise 21.20. *For the chain of Example 21.10 with* $a = 0.2$ *and* $b = 0.05$, *find* π *and the spectral gap, and compute how many steps are needed to bring* $\left\| P^k x_0 - \pi \right\|$ *below* 10^{-3} *starting from* $x_0 = (1, 0)$. *Verify your answer by computing* $P^k x_0$ *directly for that* k.

Exercise 21.21. *Let* $A \geq 0$ *be irreducible with Perron root* ρ *and Perron vector* z. *Show that* $C = \rho^{-1} \operatorname{diag}(z)^{-1} A \operatorname{diag}(z)$ *is row-stochastic, and that* A *and* C *have the same eigenvalues up to the factor* ρ. *Explain why this reduces every question about the spectrum of a nonnegative matrix to a question about a stochastic one.*

Exercise 21.22. *Prove Proposition 21.11 again for the case* $\lambda = 1$: *show that the eigenvalue* 1 *of* G *is simple without invoking Theorem 21.5, assuming* $v > 0$. *Then compute the number of power iterations needed for* 10^{-6} *accuracy at* $\alpha = 0.5, 0.85$ *and* 0.95, *and say what is being traded.*

Exercise 21.23. *Let* $A \geq 0$ *with* $A^{t} A$ *irreducible. Prove that the leading right singular vector may be chosen nonnegative, and that every later one has entries of both signs. Then verify that the matrix* M *of Section 21.6 has rank* 3, *and explain why the expansion* $M = \sum_{i=1}^{3} \sigma_i u_i v_i^{t}$ *is not a nonnegative factorisation even though it reproduces* M *exactly.*

References and further reading

Berman and Plemmons [65] is the standard monograph on nonnegative matrices and covers the whole of this chapter, including the parts of Frobenius' theory about the period that we only stated: the fact that an irreducible matrix with h eigenvalues of modulus ρ has them equally spaced around the circle, and that h is the greatest common divisor of its cycle lengths, is due to Frobenius, with the sharp bound on when $A^k > 0$ due to Wielandt. Horn and Johnson [335] give a shorter treatment and several alternative proofs of Theorem 21.5; the one used here, through the maximisation in part (d), is Wielandt's. Banica [51], our baseline, devotes a part of its own to positive matrices in this entrywise sense and is worth reading alongside for the connections to graph theory. For PageRank, Langville and Meyer [400] is a whole book about Proposition 21.11 and its consequences, and is unusually good on what happens when the theory meets a matrix with billions of rows. Nonnegative factorisation entered general use through Lee and Seung [413]; its NP-hardness is due to Vavasis [626], and the nonnegative rank of the matrix in Section 21.6 is from Cohen and Rothblum [152].